ENMATECH

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PRODUCT DATASHEET · POWER SYSTEM TRAINING SIMULATOR

Karuma HPP Digital Twin

Hybrid power system simulator – hydro, solar PV, wind, battery storage, diesel & grid – with integrated SCADA, protection relays, 3D training lab and instructor station



Real-time digital twin · Newton-Raphson load flow · Electromechanical dynamics · Hydro · Solar · Wind · BESS · Diesel · 69 protection functions · SCADA HMI · 3D training lab · Instructor & trainee workstations · Windows 10/11 · .exe

ASFAN

Software designed & developed by ASFAN

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Datasheet DS-KHPP-DT-01
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Software version 1.0.0

Product overview

The **Karuma HPP Digital Twin** is a complete, real-time software replica of the Karuma hydro power plant training system. It reproduces the entire power chain — **prime mover, synchronous generator, step-up transformer, 220 kV transmission lines, substation switchgear & protection, distribution transformers with voltage regulation and distribution loads** — together with a **solar PV plant, a wind farm, a battery energy storage system and a diesel power station** — and presents it through a professional SCADA interface, protection relay panels, an interactive 3D training lab and a dedicated instructor station.

Trainees **monitor V, I, MW, MVAr and Hz**, **control breakers, loads and generation**, **analyse power flows** and **run training scenarios and fault conditions** safely and repeatably. Every quantity is calculated by a physics-based network and machine model, not by pre-recorded data.

FREQUENCY	TOTAL GENERATION	TOTAL LOAD	VOLTAGE (HV)
50.00 Hz	120 MW	118.5 MW	220 kV

Key features

- Power chain: turbine, generator, step-up transformer, 220 kV lines, substation, distribution transformers and loads
- Hybrid energy sources: hydro, solar PV (30 MWp), wind (24 MW), battery storage (20 MW / 40 MWh), diesel (12 MW) and national grid, with live energy mix and renewable share
- SCADA system: Overview, Generation, Energy Sources, Transmission, Distribution, Alarms, Reports and Trends pages
- Training room set-up: one instructor workstation and multiple trainee workstations over the LAN; in the 3D lab the video wall shows all energy sources and each trainee desk is dedicated to one energy type
- Real-time electromechanical simulation with AVR, governor, AGC and tap-changer control
- Short-circuit faults of any type at any location, cleared by realistic numerical protection

Typical applications

- Operator training for hydro power plants, transmission and distribution utilities
- University and technical-college power system laboratories
- Protection engineering courses (distance, differential, overcurrent, UFLS, auto-reclose)
- Black-start, islanding and grid-synchronisation drills
- Renewable integration: variability, storage dispatch and diesel back-up
- Trainee assessment with automatic scoring and printable reports

System architecture



SIMULATION CORE
Node-breaker topology processor · Newton-Raphson load flow per island · machine, turbine, PV, wind, BESS & diesel dynamics
Symmetrical-component fault analysis · protection relay engine · SCADA historian · alarms & SOE
Scenario assessment engine · classroom network server (instructor ⇄ trainee workstations)

SCADA – Power system overview

The SCADA overview shows the complete single-line diagram from the generator terminals to the customer loads, with live, colour-coded voltage levels, animated power flow, breaker states and the four key system indicators requested by the operator: **Frequency, Total Generation, Total Load and Voltage (HV)**. Any breaker can be operated directly from the diagram (with interlocking and synchro-check); clicking an equipment item opens its detailed page.

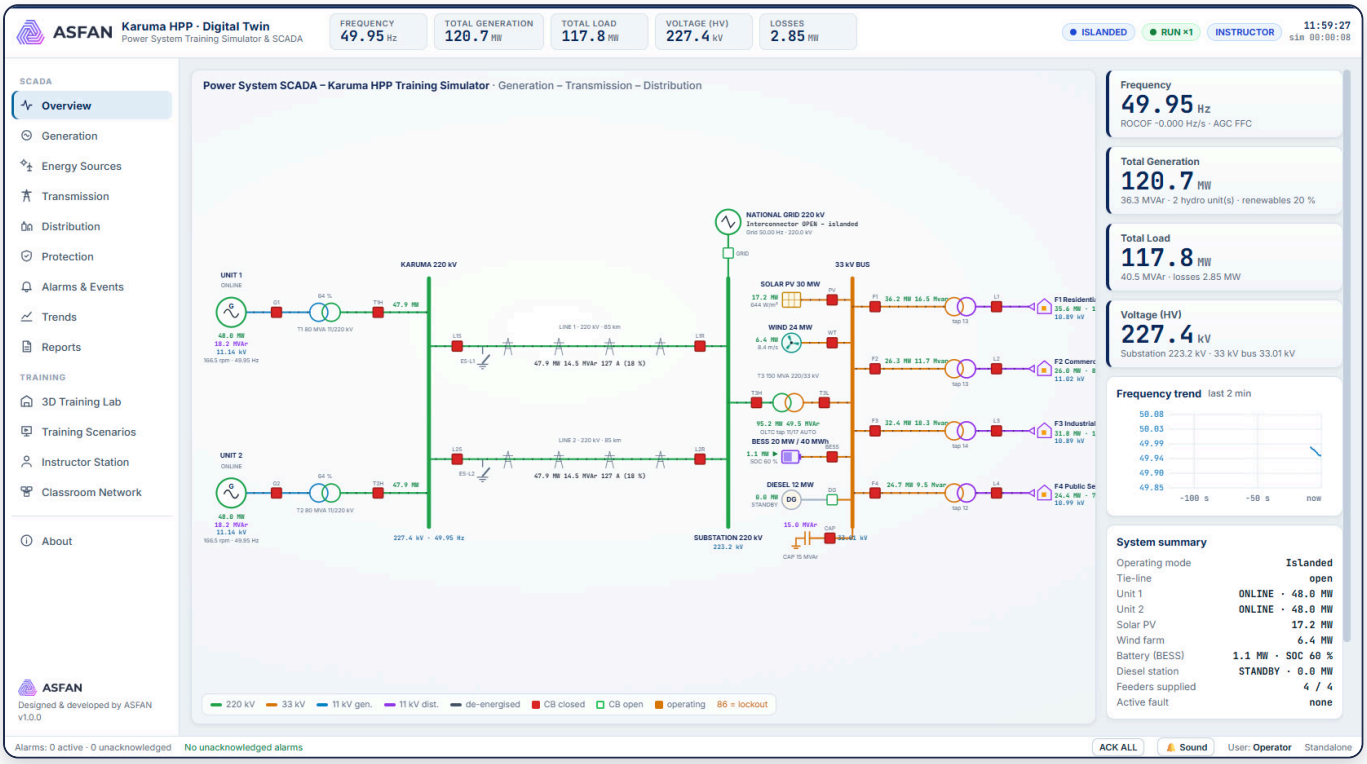


Figure 1 – SCADA overview: single-line diagram with all energy sources, key indicators, frequency trend, system summary and active alarms.

Simulated network

Section	Equipment represented
Generation	2 × 75 MVA, 11 kV, 0.85 pf hydro generators (63.75 MW each), Francis turbines with penstock, governors, static excitation & AVR, generator circuit breakers
Step-up	2 × 80 MVA, 11/231 kV Dyn11 step-up transformers with off-circuit taps, 220 kV breakers, Karuma 220 kV busbar
Transmission	220 kV double-circuit overhead line, 2 × 85 km, line breakers at both ends, line earthing switches, 2 500 MVA national-grid interconnector
Substation	Substation 220 kV busbar, 150 MVA 220/33 kV YNyn0 power transformer with on-load tap changer (±8 × 1.25 %), 33 kV busbar, 15 MVAr capacitor bank
Distribution	4 × 33 kV feeders (4–6 km), 4 distribution transformers 31.5–50 MVA 33/11 kV with OLTC & automatic voltage control, 11 kV incomers
Energy sources	Solar PV 30 MWp, wind farm 6 × 4 MW, BESS 20 MW / 40 MWh and diesel station 3 × 4 MW, each on a 33 kV collector cable with breaker and protection
Loads	Residential (36 MW), Commercial (26 MW), Industrial with motor load (32 MW), Public services & hospital (24.5 MW) – voltage- and frequency-dependent

Generation – Generator Control & AVR

Each unit has a full control faceplate reproducing the **Generator Control & AVR panel**: start / stop / emergency-stop sequences, speed and load set-points, governor mode (droop or isochronous), field breaker and AVR (auto / manual), synchroscope with manual and automatic synchronising, P–Q capability diagram and all turbine and generator measurements.

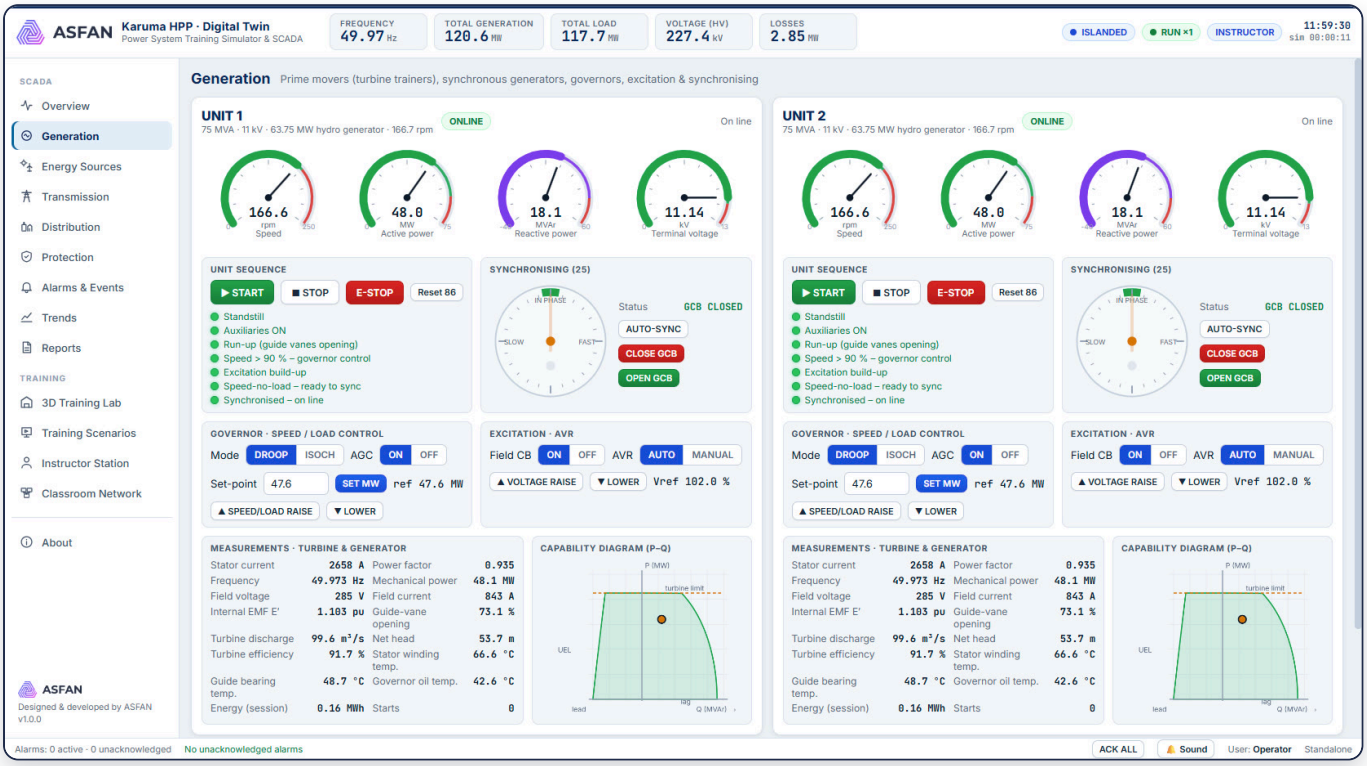


Figure 2 – Generation page with two unit faceplates, gauges, start sequence, synchroscope and capability diagram.

Synchronous generator & turbine (per unit)	
Rating	75 MVA, 11 kV, pf 0.85, 63.75 MW
Speed	166.7 rpm, 36 poles, 50 Hz
Reactances	Xd 1.05 · Xd' 0.30 · Xd'' 0.22 · X2 0.22 pu
Time constants	T'd0 6 s · H 4.0 s
Prime mover	Francis turbine, 54 m net head, water-hammer (Tw) model, guide-vane servo with rate limits
Measurements	V, I, MW, MVAr, pf, Hz, rpm, gate %, discharge, head, efficiency, field V/I, temperatures, energy

Controls	
Governor	PI speed/load controller, 4 % droop or isochronous, load ramp limiter, load-reference feed-forward
Excitation / AVR	Static exciter, PI AVR with reactive droop, V/Hz limiter, manual field control, field breaker
Synchronising	Synchroscope (Δθ, Δf, ΔV), synchro-check relay 25, auto-synchroniser
AGC & network control	OFF / flat-frequency / tie-line-bias modes, frequency bias, interchange schedule, ACE display
Sequences	Auxiliaries → run-up → speed control → excitation → speed-no-load → synchronised

Energy sources – hybrid power system

Besides the two hydro units, the digital twin contains a complete set of **renewable, storage and back-up energy sources** connected to the 33 kV busbar through their own collector cables, breakers and protection: a **solar PV plant**, a **wind farm**, a **battery energy storage system (BESS)** and a **diesel power station**, plus the **national grid** interconnector. Each source is modelled with its own physics and controls, takes part in the load flow and in the frequency response, and is supervised by grid-code protection.

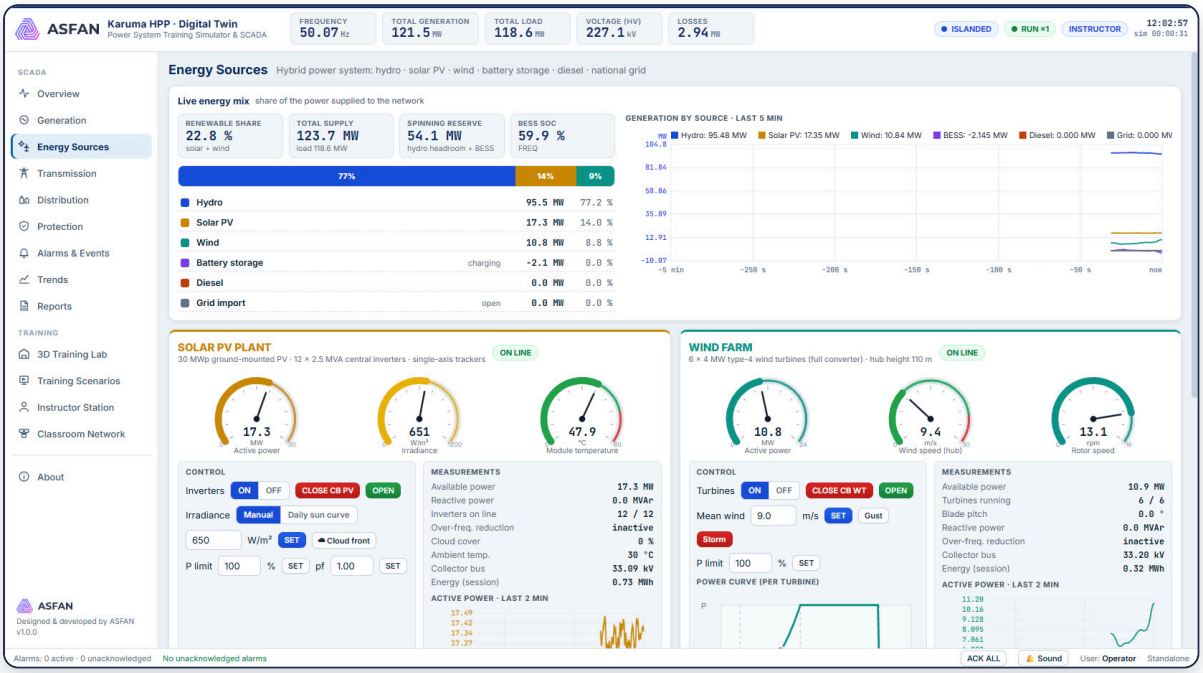


Figure 3 – Energy Sources page: live energy mix and renewable share, generation by source, and one control/measurement panel per source with gauges and trends.

Source	Rating	Model & controls
Hydropower	2 × 75 MVA, 63.75 MW Francis units	Penstock, governor (droop / isochronous), AVR; slack and frequency reference of the system
Solar PV	30 MWp · 12 × 2.5 MVA central inverters · single-axis trackers	Manual irradiance or daily sun curve, cloud fronts, module temperature, P-limit (curtailment), power factor, over-frequency power reduction (40 %/Hz above 50.2 Hz)
Wind	6 × 4 MW type-4 turbines · hub height 110 m	Turbulent wind with gusts, power curve (cut-in 3, rated 12.5, cut-out 25 m/s), rotor speed & pitch, storm shutdown and automatic restart, curtailment
Battery storage	20 MW / 40 MWh lithium-ion (LFP) · 8 containers	Modes: idle, manual set-point, frequency response (droop 40 MW/Hz), renewable smoothing, peak shaving; SOC limits 5–95 %, efficiency, cell temperature, cycle count
Diesel	3 × 4 MW gensets · 1500 rpm	Start sequence (cranking → rated speed → auto-synchronising → on load), 4 % droop, fuel consumption & day tank, loss-of-mains trip, stop / refuel
National grid	220 kV interconnector, 2 500 MVA	Import / export, synchro-check, AGC tie-line bias control

Grid-code supervision of PV, wind and BESS: disconnection on dead bus, V < 0.8 pu or f outside 47.5–52 Hz; automatic reconnection after 20 s within 0.9–1.1 pu and 49.5–50.2 Hz. Instructor weather controls: irradiance, cloud front, sun curve, mean wind, gusts and storms.

Control-room screens – one screen per energy source

In the 3D training lab the **central SCADA video wall shows all energy sources together** — hydro, solar PV, wind, battery storage, diesel and grid — with the live energy mix and renewable share. **Every trainee desk is dedicated to one energy source** and its two monitors show that source's operator screens. The same screen set can be supplied for the customer's real control room or training room: **one wall display for the complete energy mix plus dedicated monitors for each energy type**.

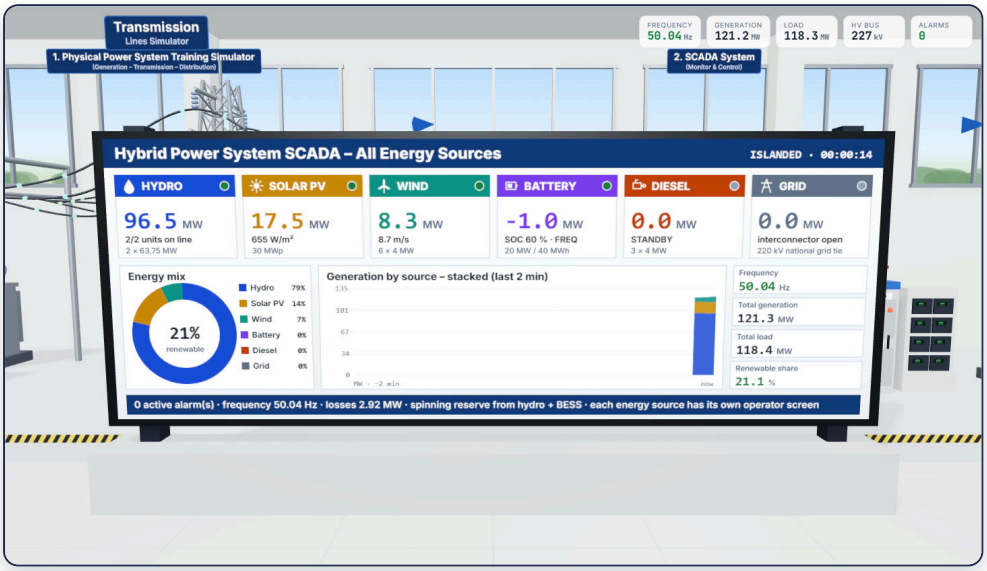


Figure 4 – SCADA video wall: all energy sources, energy-mix donut, generation by source and key indicators.



Figure 5 – Solar PV desk: plant overview and inverters & sun-path screens (readable when the camera approaches).

Desk	Monitor 1 · Monitor 2
Hydropower	Unit 1 · Unit 2 (P, Q, V, speed, gate, head, temperatures)
Solar PV	Plant overview · 12 inverters & irradiance profile
Wind	Wind farm · 6 turbines & power curve
Battery storage	SOC & power · 8 containers & P–f droop
Diesel	Station overview · 3 gensets & start sequence
Grid & mix	Interconnector & synchro-check · energy mix
Instructor	All-sources overview · alarms

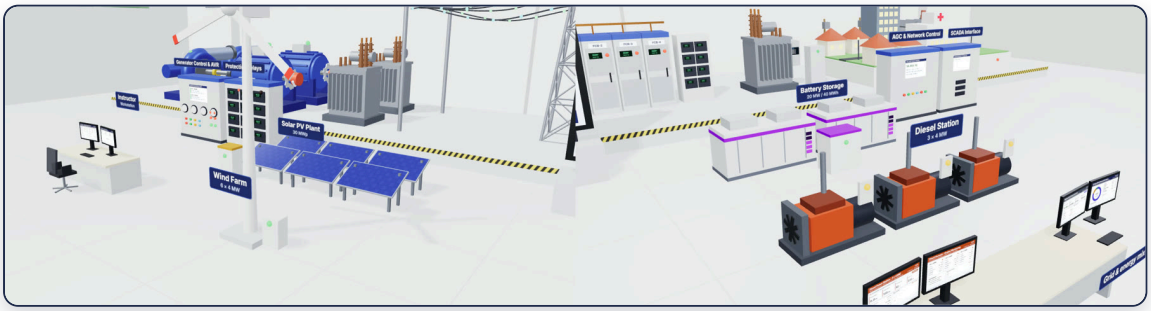


Figure 6 – 3D models in the training lab: solar PV array and wind turbine (left), battery containers with SOC indicators and diesel gensets (right); rotor, fans and status lamps follow the simulation.

Transmission & distribution

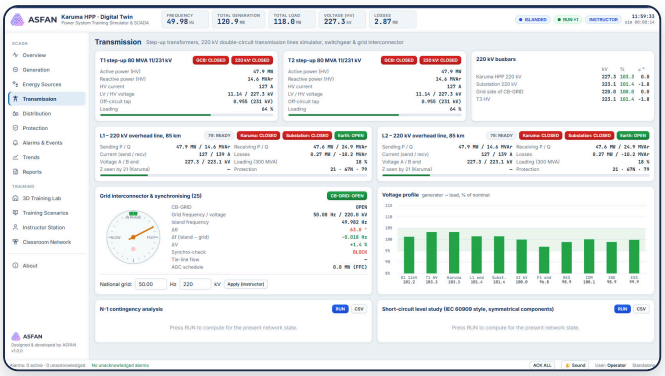


Figure 7 – Transmission: step-up transformers, 220 kV lines, grid synchronising, voltage profile, N-1 and short-circuit studies.

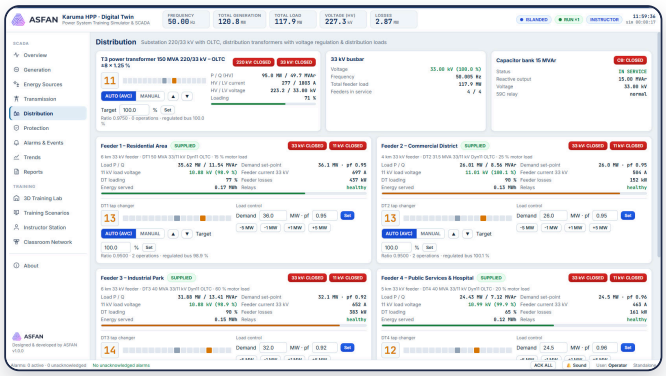


Figure 8 – Distribution: T3 OLTC, capacitor bank, four feeders with tap changers and load control.

Transmission functions

- Sending / receiving P, Q, current, losses and loading of each line and transformer
- Line parameters: Z1 = 0.06 + j0.41 Ω/km, Z0 = 0.30 + j1.15 Ω/km, shunt charging; rating 300 MVA
- Grid interconnector with synchroscope, synchro-check and grid frequency/voltage set by the instructor
- Busbar voltage magnitudes and angles; voltage profile from the generator to the load
- **N-1 contingency analysis** and **short-circuit level study** (3-phase and 1-phase, S″k, X/R) on demand, exportable to CSV

Distribution functions

- On-load tap changers on the 150 MVA substation transformer and all distribution transformers – AUTO (AVC) with dead-band and time delays, or MANUAL raise / lower
- 15 MVAr capacitor bank for reactive compensation
- Load demand and power-factor set-points, ±1 / ±5 MW load steps, 24-hour daily load profile
- Load models with voltage exponents and frequency sensitivity; motor load in the industrial park
- Energy served and energy-not-served accounting

Power flow & analysis engine

Topology	Node-breaker model with breakers and earthing switches; automatic bus and island detection
Load flow	Full Newton-Raphson AC load flow for every electrical island, distributed slack by unit inertia, recalculated every integration step
Dynamics	Swing equation per island, turbine/penstock, governor and excitation dynamics; time step 20 ms (5 ms during faults and switching)
Fault analysis	Symmetrical components (positive, negative, zero sequence) incl. transformer vector groups; 3PH, L-G, L-L, L-L-G faults with fault resistance at any bus or any % of a line

Protection relays

The protection system contains **69 numerical relay functions** (IEC 60255 / ANSI C37.2) evaluated in real time from simulated CT and VT signals. Relays start, time and trip breakers exactly like real devices, including breaker-failure back-up, lock-out (86) and auto-reclose. Each fault automatically creates a **disturbance record** with three-phase current and voltage waveforms and the trip sequence.

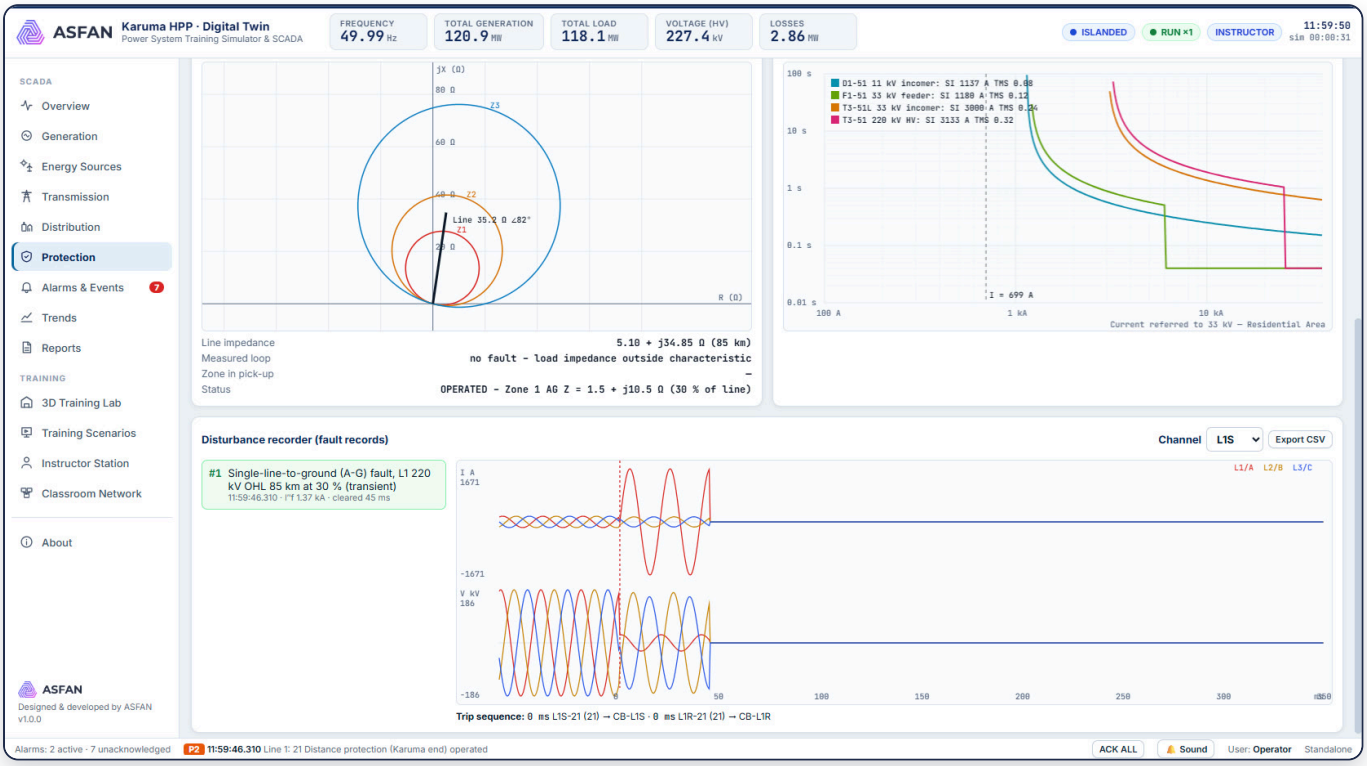


Figure 9 – Distance-protection R–X diagram, time–current coordination chart and disturbance recorder after a single-phase line fault cleared in 45 ms.

Protected zone	Functions (ANSI)
Generators	**87G** differential · **64G** stator earth fault · **51V** voltage-restrained O/C · **46** negative sequence · **32** reverse power · **40** loss of excitation · **81U/81O** · **59/27**
Transformers	**87T** differential · **50/51** phase O/C (IEC SI/VI/EI curves) · **51N** earth fault · 33 kV and 11 kV incomer O/C
Busbars	**87B** busbar differential (Karuma 220 kV, substation 220 kV, 33 kV)
220 kV lines	**21** distance (mho, Z1 80 %, Z2 120 %, Z3 220 %, POTT, six loops with k0) · **67N** directional E/F · **79** auto-reclose with lock-out
Grid tie	**67** directional O/C · **81R/78** islanding (ROCOF) · **25** synchro-check
Distribution	Feeder **50/51**, **50N/51N** · DT **87T** · 3-stage under-frequency load shedding **81U** · motor **27** · capacitor **59C**
Energy sources	**50/51** collector-cable O/C for PV, wind, BESS and diesel · grid-code **27/59/81** disconnection & reconnection · diesel loss-of-mains · 33 kV **87B** trips the source breakers
All breakers	**50BF** breaker-failure back-up tripping · **86** lock-out with reset · switching interlocks · earthing-switch interlocks

SCADA – Alarms, trends & reports

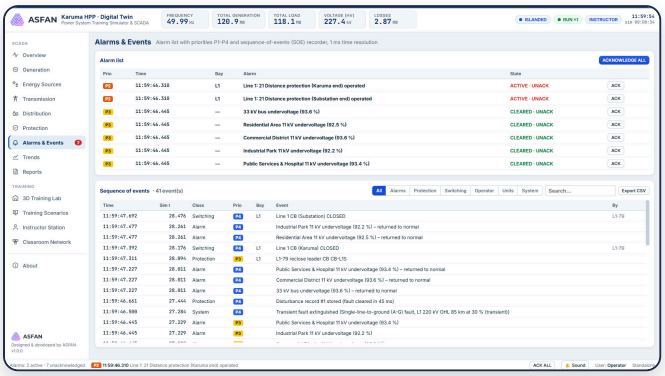


Figure 10 – Alarm list (P1–P4) and sequence-of-events recorder.

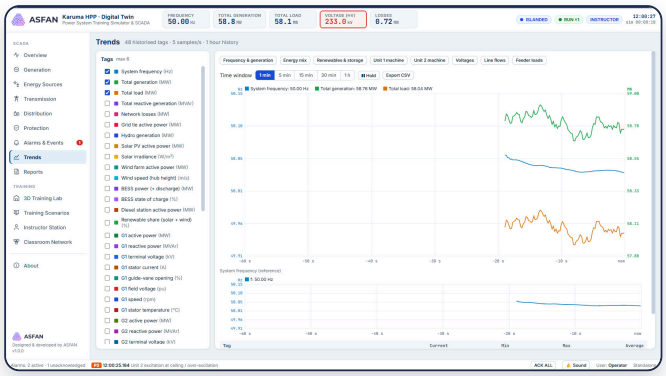


Figure 11 – Trends: 48 historised tags, up to 6 pens, 1 min – 1 h windows.

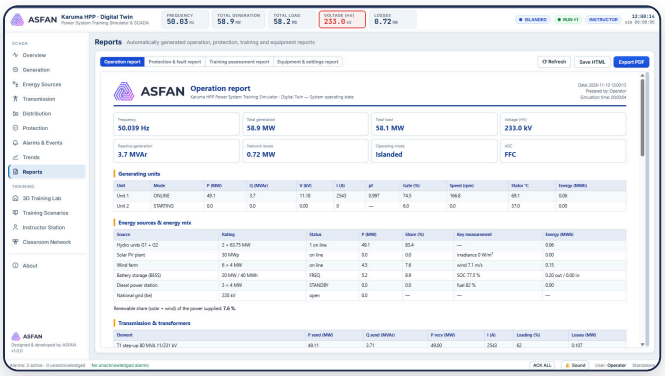


Figure 12 – Automatic reports with PDF export.

Alarms & events

- Four alarm priorities, audible alarm, acknowledge single / all, alarm ticker
- Sequence of events with 1 ms resolution: switching, protection, operator, unit and system events; filter, search and CSV export

Trends

- 1 hour history at 5 samples per second, hold / resume, statistics (min / max / average), CSV export

Reports

- Operation report · Protection & fault report · Training assessment report · Equipment & settings report – exported as PDF or HTML

Training lab, scenarios & instructor station



Figure 13 – Trainee workstations: one desk per energy source (hydro, solar PV, wind, battery, diesel, grid & mix) – shown: the hydropower desk; click a monitor to zoom in.

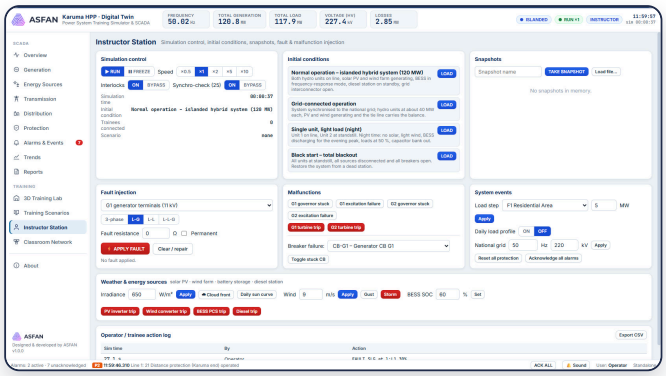


Figure 14 – Instructor station: run / freeze / speed, initial conditions, snapshots, faults and malfunctions.

10 training scenarios with automatic assessment

Scenario	Level	Duration
Unit start-up & manual synchronisation	BASIC	12 min
Frequency control after a load increase	BASIC	8 min
220 kV line fault – isolation & restoration	INTERMEDIATE	15 min
Voltage regulation with OLTC, capacitor bank & AVR	INTERMEDIATE	10 min
Synchronisation to the national grid & tie-line control	INTERMEDIATE	10 min
Generator trip, under-frequency load shedding & recovery	ADVANCED	15 min
Black start & system restoration	ADVANCED	25 min
Feeder fault with breaker failure (50BF)	ADVANCED	12 min
Renewable variability – cloud front & wind storm	INTERMEDIATE	10 min
Evening peak with one hydro unit – storage & diesel dispatch	ADVANCED	12 min

Instructor station

- 4 initial conditions: islanded hybrid system, grid-connected, single unit / evening, total blackout
- Weather & sources: irradiance, cloud front, sun curve, wind, gusts, storm, BESS SOC, PV / wind / BESS / diesel trips
- Snapshots save / restore (memory and file)
- Fault injection: 3PH, L-G, L-L, L-L-G, fault resistance, transient or permanent, any bus or line position
- Malfunctions: governor stuck, excitation failure, turbine trip, stuck breaker; interlock and synchro-check bypass

Classroom network

- Instructor workstation hosts the simulation; trainee workstations join over the training-room LAN
- All trainees see the same live plant and operate it from their own SCADA screens
- Instructor-only commands are protected; each trainee action is logged by name
- Scores, objectives, penalties and quiz answers stored in the training assessment report

Technical specifications

Software	
Product	Karuma HPP Digital Twin – Hybrid Power System Training Simulator & SCADA, version 1.0.0
Delivery	Windows installer (Setup .exe) and portable .exe – no additional runtime required
Operating modes	Standalone · Instructor workstation (host) · Trainee workstation (client)
Simulation	Real time with ×0.25 – ×10 speed and freeze; 20 ms step (5 ms during transients)
Data	48 historised tags (1 h, 5 Hz), 5 000-event SOE buffer, 20 disturbance records
Exports	PDF and HTML reports; CSV for events, trends, studies, disturbance records and action log; JSON snapshots
Network	TCP/IP LAN, HTTP + Server-Sent Events, default port 47820
Language / units	English user interface, SI units, IEC / ANSI device numbering

Recommended workstation	
Operating system	Windows 10 or Windows 11, 64-bit
Processor / memory	4-core CPU, 2.5 GHz or faster · 8 GB RAM (16 GB recommended for the instructor workstation)
Graphics	DirectX 11 / WebGL 2 capable GPU for the 3D training lab
Display	1920 × 1080 or higher (dual monitors recommended for the instructor)
Disk	500 MB free space

Scope of supply	
Software	Karuma HPP Digital Twin installer and portable executable
Documentation	This product datasheet (included in the installation)
Control-room screens (optional)	Video-wall display for the complete energy mix and dedicated operator monitors per energy source (hydro, solar PV, wind, battery storage, diesel, grid)
Services (optional)	Installation and commissioning, instructor training, additional customer-specific scenarios, model adaptation to site data

ASFAN

Software developer: ASFAN

The Karuma HPP Digital Twin software — simulation models, energy sources, SCADA, protection, 3D training lab and instructor tools — is **designed and developed by ASFAN**.

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